A single NAND axiom for Boolean algebra

Abstract

We give an explicit proof that the equation

$$((a * b) * c) * \big(a * ((a * c) * a)\big) = c$$

characterizes the NAND reducts of Boolean algebras on nonempty carriers. The proof has two parts: a sequence of substitutions establishes commutativity and double negation; an order-theoretic construction then recovers the Boolean lattice. We also explain why the equation is complete for identities in the NAND language.

1 Statement and plan

Let A be a nonempty set equipped with a binary operation $*$ satisfying

$$\boxed{((x * y) * z) * \big(x * ((x * z) * x)\big) = z} \tag{W}$$

for all $x, y, z \in A$. No other property of $*$ is assumed. In particular, parentheses must be retained: even in a Boolean algebra, NAND is not associative.

Theorem 1. *There is a Boolean algebra structure on A whose NAND operation is $*$. Writing $n(x) = x * x$, its operations are*

$$\neg x = n(x), \qquad x \wedge y = n(x * y), \qquad x \vee y = n(x) * n(y),$$

and its constants are

$$\mathbf{1} = x * n(x), \qquad \mathbf{0} = n(\mathbf{1}).$$

The expression defining $\mathbf{1}$ is independent of x. Conversely, NAND in any Boolean algebra satisfies (W)*.*

We include the one-element Boolean algebra. Nonemptiness is needed to obtain the constants: an empty algebra satisfies (W) vacuously.

The difficult part is the following purely equational lemma. Its proof occupies Section 2. Section 3 explains how its identities determine a bounded distributive lattice with complement, and Section 4 proves the converse and completeness.

Lemma 2 (The algebraic lemma)**.** *Put $n(x) = x * x$. The single identity* (W) *implies*

$$x * y = y * x, \tag{C}$$
$$n(n(x)) = x, \tag{DN}$$
$$n(x) * (x * y) = x, \tag{I}$$
$$(x * y) * (y * (x * z)) = y, \tag{S}$$
$$x * (y * (x * (y * z))) = x * (y * z). \tag{J}$$

2 From the single axiom to the algebraic lemma

The repeated right-hand factor in (W) suggests introducing

$$q(x,y) = x * ((x * y) * x), \qquad m(x) = q(x,x).$$

Then the axiom reads

$$((x * y) * z) * q(x,z) = z.$$

We write $\mathrm{W}(x,y,z)$ for this instance of the axiom. Our intermediate goals are $m(x) = x * x$ and $q(x,y) = y * x$. These simplify the repeated expressions enough to reveal double negation and commutativity. Until commutativity has been proved, the order of every pair of factors is significant.

2.1 Four substitution identities

The instance $\mathrm{W}(x, (x * y) * x, y)$ gives

$$(q(x,y) * y) * q(x,y) = y. \tag{R}$$

Expanding the outer q and applying (R), we obtain

$$q(q(x,y),y) = q(x,y) * y. \tag{R'}$$

For the next identity, use $\mathrm{W}((v * w) * x, q(v,x), y)$. Its left-hand side is

$$\Big(\big(((v * w) * x) * q(v,x)\big) * y\Big) * q((v * w) * x, y).$$

The parenthesized factor equals x by $\mathrm{W}(v,w,x)$, so

$$(x * y) * q((v * w) * x, y) = y.$$

Every element z can be written as a product, since $\mathrm{W}(z,z,z)$ gives $z = ((z * z) * z) * m(z)$. Thus $v * w$ in the preceding identity can be any $z \in A$, and

$$(x * y) * q(z * x, y) = y. \tag{C_0}$$

In (C_0), substitute $x = q(a,b)$, $y = c$, and $z = (a * a) * b$. The instance $\mathrm{W}(a,a,b)$ simplifies $z * x$ to b. Hence

$$(q(a,b) * c) * q(b,c) = c. \tag{D_0}$$

Finally, substitute $(a,b,c) = (x, q(y,z), z)$ into (D_0) and use (R'):

$$(q(x, q(y,z)) * z) * (q(y,z) * z) = z. \tag{J_0}$$

2.2 Identifying the square and proving double negation

Fix $a \in A$ and abbreviate

$$u = m(a), \qquad t = (a * a) * a.$$

The definition of m and $\mathrm{W}(a,a,a)$ give

$$a * t = u, \qquad t * u = a. \tag{1}$$

We first derive four relations involving a and u:

$$(a * u) * a = u, \tag{U_1}$$
$$u * u = a, \tag{U_2}$$
$$u * (u * a) = a, \tag{U_3}$$
$$a * (u * (a * u)) = u. \tag{U_4}$$

To prove (U_1), apply (C_0) with $(x, y, z) = (a, u, a * a)$. This gives $(a * u) * q(t, u) = u$, while

$$q(t, u) = t * ((t * u) * t) = t * (a * t) = t * u = a.$$

Now $\mathrm{W}(a, u, a)$ becomes $((a * u) * a) * u = a$; applying (U_1) proves (U_2).

For (U_3), note that $q(a, u) = a * ((a * u) * a) = a * u$. In ($J_0$), set $x = y = z = a$. Since $q(a, a) = u$, the result is

$$((a * u) * a) * (u * a) = a,$$

which reduces to (U_3). Finally, (C_0) with $(x, y, z) = (t, u, a)$ gives $a * q(u, u) = u$. Expanding $q(u, u)$ and using $u * u = a$ proves (U_4).

We need one identity that applies these relations at a different element. For arbitrary d, b, set

$$v = (d * d) * d, \qquad r = q(b * (d * d), d).$$

By (C_0), $v * r = d$. Therefore

$$q(v, r) = v * ((v * r) * v) = v * (d * v) = v * m(d) = d,$$

where the last equality is $\mathrm{W}(d, d, d)$. Now (J_0) at $(x, y, z) = (v, b * (d * d), d)$ yields

$$(d * d) * (q(b * (d * d), d) * d) = d.$$

Replace d by $m(a)$ and use (U_2), which holds for every a:

$$a * (q(b * a, m(a)) * m(a)) = m(a). \tag{V}$$

Return to our fixed a and put $w = u * (a * u)$. By (U_4) and (U_2), respectively,

$$a * w = u, \qquad m(u) = u * ((u * u) * u) = w. \tag{2}$$

The instance $\mathrm{W}(u, u, u * a)$ reduces, using (U_2) and (U_3), to

$$(a * (u * a)) * w = u * a. \tag{3}$$

Apply (V) with (a, b) replaced by (u, t). Since $t * u = a$ and $m(u) = w$, this gives

$$u * (q(a, w) * w) = w.$$

But $q(a, w) = a * ((a * w) * a) = a * (u * a)$. By (3), the last displayed left-hand side is $u * (u * a) = a$. Thus $w = a$, and $a * w = u$ now gives

$$\boxed{m(a) = a * a.} \tag{4}$$

Substituting this into $u * u = a$ proves double negation (DN). We also record two consequences needed below:

$$(a * a) * (a * (a * a)) = a, \tag{H}$$
$$((a * b) * a) * (a * a) = a. \tag{L}$$

The first is $w = a$ after substituting $u = a * a$; the second is $\mathrm{W}(a, b, a)$ after substituting $m(a) = a * a$.

2.3 Identifying the reversed product

We next prepare identities that let us compare $q(a,b)$ with $b*a$. First, $\mathrm{W}(q(a,b),c,b)$ and (R') give

$$((q(a,b)*c)*b)*(q(a,b)*b) = b. \tag{P_0}$$

Apply (C_0) with

$$x = q(a,b)*b, \qquad y = c, \qquad z = (q(a,b)*b)*b.$$

By (P_0) at $c = b$, the product $z*x$ is b, so

$$((q(a,b)*b)*c)*q(b,c) = c. \tag{E_0}$$

In this identity replace (a,b,c) by $(b,a,q(b,a))$. Its first factor is a by (R), giving

$$a*q(a,q(b,a)) = q(b,a). \tag{K_0}$$

Two more substitutions into the original axiom yield

$$a*q(b*c,q(b,a)) = q(b,a), \tag{B_0}$$
$$a*q(b,q(c*b,a)) = q(c*b,a). \tag{I_0}$$

For (B_0), use $\mathrm{W}(b*c,a,q(b,a))$ and simplify the first factor by $\mathrm{W}(b,c,a)$. For (I_0), use $\mathrm{W}(b,a,q(c*b,a))$ and simplify the first factor by (C_0).

We first identify q on pairs whose second entry is a square:

$$\boxed{q(b*a,a*a) = a.} \tag{T_0}$$

Here are the intermediate steps. Let $r = q(x*y,y)$. Equation (C_0) gives $(y*y)*r = y$, and hence

$$q(y*y,r) = (y*y)*(y*(y*y)) = y$$

by (H). In (I_0), take $(a,b,c) = (r,y,y)$ to get

$$r*q(y,q(y*y,r)) = q(y*y,r).$$

This reduces to $r*q(y,y) = y$. Since $q(y,y) = m(y) = y*y$,

$$q(x*y,y)*(y*y) = y. \tag{N_0}$$

Now apply the original axiom at

$$\mathrm{W}\big(b*a,\ ((b*a)*a)*(b*a),\ a*a\big).$$

The first factor is $q(b*a,a)*(a*a) = a$ by (N_0), leaving

$$a*q(b*a,a*a) = a*a. \tag{O_0}$$

Put $s = q(b*a,a*a)$. Equation (I_0) at $(a*a,a,b)$ says $(a*a)*q(a,s) = s$. By (O_0), $a*s = a*a$, so

$$q(a,s) = a*((a*s)*a) = a*((a*a)*a) = m(a) = a*a.$$

Thus $s = (a*a)*(a*a) = a$, proving (T_0).

2.4 Commutativity

Fix a, b and write

$$t = (a * b) * a, \qquad r = q(a, b) = a * t.$$

The next three equalities relate a, t, r:

$$t * r = a, \qquad r * a = t, \qquad t * a = r. \tag{5}$$

Indeed, (T_0) and (L) give $q(t, a * a) = a$ and $t * (a * a) = a$. Expanding the former,

$$a = q(t, a * a) = t * ((t * (a * a)) * t) = t * (a * t) = t * r.$$

This proves the universal identity

$$((x * y) * x) * q(x, y) = x.$$

Apply it with $(x, y) = (t, a * a)$. The two known simplifications give $(a * t) * a = t$, or $r * a = t$. Finally,

$$q(a, t) = a * ((a * t) * a) = a * (r * a) = a * t = r,$$

and

$$q(t, r) = t * ((t * r) * t) = t * (a * t) = t * r = a.$$

Equation (K_0) with $(a, b) = (t, a)$ is therefore $t * a = r$. This proves all of (5).

We now establish

$$(a * a) * q(a, b) = a. \tag{W_1}$$

For an arbitrary c, (C_0) gives $(a * a) * q(c * a, a) = a$. Consequently,

$$q(a * a, q(c * a, a)) = (a * a) * (a * (a * a)) = a$$

by (H). Thus (J_0) at $(x, y, z) = (a * a, c * a, a)$ becomes

$$(a * a) * (q(c * a, a) * a) = a. \tag{6}$$

Take $c = t$. Using (5),

$$q(t * a, a) = q(r, a) = r * ((r * a) * r) = r * (t * r) = r * a = t.$$

Equation (6) becomes $(a * a) * (t * a) = a$; since $t * a = r$, this is (W_1).

We can now identify q everywhere. Equation (B_0) at $c = b$ says

$$a * q(b * b, q(b, a)) = q(b, a).$$

Its inner expression simplifies by (W_1) and (H):

$$\begin{aligned} q(b * b, q(b, a)) &= (b * b) * (((b * b) * q(b, a)) * (b * b)) \\ &= (b * b) * (b * (b * b)) = b. \end{aligned}$$

We have proved

$$\boxed{q(b, a) = a * b.} \tag{Q_1}$$

Finally, rewrite (K_0) using (Q_1):

$$\begin{aligned} q(b, a) &= a * q(a, q(b, a)) \\ &= a * (q(b, a) * a) \\ &= a * ((a * b) * a) \\ &= q(a, b). \end{aligned}$$

The first and last expressions are $a * b$ and $b * a$, respectively. This proves commutativity (C).

2.5 The remaining three identities

We may now use commutativity freely. Equation (W_1), together with (Q_1), gives $n(x)*(x*y) = x$, proving (I). The original axiom and (Q_1) give

$$((x * y) * z) * (x * z) = z.$$

Replacing (y, z) by (z, y) and commuting the indicated factors yields $(x * y) * (y * (x * z)) = y$, which is (S). Finally, (I_0) and (Q_1) give

$$a * ((a * (c * b)) * b) = a * (c * b).$$

Commutativity turns this into (J). Together with double negation, this finishes the proof of Lemma 2. □

3 Recovering the Boolean lattice

From this point onward we use only the five identities in Lemma 2. In particular, $*$ is commutative and n is an involution, so n is both injective and surjective.

3.1 Absorption identities and constants

We begin with three useful consequences:

$$x * (x * y) = x * n(y), \tag{B}$$
$$x * (n(x) * y) = n(x), \tag{I'}$$
$$x * n(x * y) = x * y. \tag{A}$$

For (B), put $p = x * y$. Identity (J) at (x, p, y) is

$$x * (p * (x * (p * y))) = x * (p * y).$$

Identity (S) at (y, x, p) gives $p * (x * (p * y)) = x$. Consequently $x * (y * (x * y)) = n(x)$. Use this last equality with x, y interchanged to simplify (J) at (x, y, x):

$$x * n(y) = x * (x * y).$$

This is (B).

Replacing x by $n(x)$ in (I) proves (I'). For (A), apply (I) at $(x * y, n(x))$. Its inner product is $(x * y) * n(x) = x$ by (I) and commutativity, so $n(x * y) * x = x * y$, as required.

Let $e(x) = x * n(x)$. For $d = x * y$, the preceding identities give

$$\begin{aligned} e(x) &= n(x) * x \\ &= n(x) * (n(x) * d) && \text{by (I)} \\ &= n(x) * n(d) && \text{by (B)} \\ &= n(d) * (n(d) * x) && \text{by (B) and commutativity} \\ &= n(d) * d && \text{by (A) and commutativity} \\ &= e(d). \end{aligned}$$

Interchanging x, y gives $e(y) = e(d)$ as well. Thus $e(x)$ is constant, and nonemptiness allows us to define

$$\mathbf{1} = e(x), \qquad \mathbf{0} = n(\mathbf{1}).$$

Equation (B) and double negation show that

$$x * \mathbf{1} = x * (x * n(x)) = x * n(n(x)) = n(x). \tag{7}$$

Applying (I) at $(\mathbf{1}, x)$ gives $\mathbf{0} * (\mathbf{1} * x) = \mathbf{1}$, hence $\mathbf{0} * n(x) = \mathbf{1}$. Since n is surjective,

$$\mathbf{0} * x = \mathbf{1} \qquad \text{for all } x. \tag{8}$$

3.2 The order and its reversal by negation

Define a relation on A by

$$x \le y \quad \Longleftrightarrow \quad x * y = n(x). \tag{9}$$

It is reflexive by the definition of n. To see antisymmetry, suppose $x \le y$ and $y \le x$. Commutativity gives $n(x) = x * y = y * x = n(y)$, and injectivity of n gives $x = y$.

A second description of the relation will be useful:

$$x \le y \quad \Longleftrightarrow \quad x * n(y) = \mathbf{1}. \tag{10}$$

If $x * y = n(x)$, then (B) gives $x * n(y) = x * (x * y) = x * n(x) = \mathbf{1}$. Conversely, if $x * n(y) = \mathbf{1}$, apply (B) at $(x, n(y))$:

$$x * y = x * (x * n(y)) = x * \mathbf{1} = n(x).$$

For transitivity, suppose $x \le y$ and $y \le z$. Then $y * x = n(x)$ and $y * n(z) = \mathbf{1}$. Identity (S) at $(y, n(z), x)$,

$$(y * n(z)) * (n(z) * (y * x)) = n(z),$$

therefore becomes $\mathbf{1} * (n(z) * n(x)) = n(z)$. By (7) this says $n(n(z) * n(x)) = n(z)$. Apply n to both sides to obtain $n(x) * n(z) = z$. Now (I') at $(x, n(z))$ gives

$$x * z = x * (n(x) * n(z)) = n(x),$$

so $x \le z$. This proves that (9) is a partial order.

Negation reverses it. Indeed, if $x \le y$, then

$$n(y) * n(x) = n(y) * (y * x) = y = n(n(y))$$

by (I), so

$$x \le y \quad \Longrightarrow \quad n(y) \le n(x). \tag{11}$$

Since n is involutive, this implication is an equivalence.

3.3 Meet and join

Define $x \wedge y = n(x * y)$. By (A),

$$n(x * y) * x = x * y = n(n(x * y)).$$

Thus $x \wedge y \le x$, and commutativity also gives $x \wedge y \le y$.

We must prove that this lower bound is the greatest one. Suppose $z \le a$ and $z \le b$, and put

$$k = n(z), \qquad p = a * b, \qquad h = k * p.$$

The hypotheses say $a * z = b * z = k$. Applying (S) at (a, t, z) and (b, t, z) gives, for every t,

$$(t * a) * (t * k) = t, \qquad (t * b) * (t * k) = t. \tag{12}$$

The second equality at $t = a$ is $p * (a * k) = a$. Identity (S) at (k, p, a) now gives

$$h * a = (k * p) * (p * (k * a)) = p.$$

Use the first equality of (12) at $t = h$ to calculate

$$\begin{aligned} h &= (h * a) * (h * k) \\ &= p * (k * (k * p)) \\ &= p * (k * n(p)) && \text{by (B)} \\ &= n(p) && \text{by (I') and commutativity.} \end{aligned}$$

Finally, (I) at (k, p) is $n(k) * (k * p) = k$. Since $n(k) = z$ and $k * p = h = n(p)$, it says

$$z * n(p) = n(z).$$

This is exactly $z \leq n(p) = a \wedge b$. Hence $\wedge$ is the greatest-lower-bound operation. Its commutativity, associativity, idempotence, and monotonicity follow from this universal property.

An order-reversing involution turns greatest lower bounds into least upper bounds. Explicitly, define

$$x \vee y = n\big(n(x) \wedge n(y)\big) = n(x) * n(y).$$

Since $n(x) \wedge n(y) \leq n(x), n(y)$, applying n shows $x, y \leq x \vee y$. If $x, y \leq z$, then $n(z) \leq n(x), n(y)$, so $n(z) \leq n(x) \wedge n(y)$; applying n gives $x \vee y \leq z$. Thus $\vee$ is the least-upper-bound operation, and we have a lattice. The same construction gives the De Morgan laws

$$n(x \wedge y) = n(x) \vee n(y), \qquad n(x \vee y) = n(x) \wedge n(y). \tag{13}$$

By (7), $x \leq \mathbf{1}$. By (8) and double negation, $\mathbf{0} * x = \mathbf{1} = n(\mathbf{0})$, so $\mathbf{0} \leq x$. Finally,

$$x \wedge n(x) = n(x * n(x)) = \mathbf{0}, \qquad x \vee n(x) = n(x) * x = \mathbf{1}. \tag{14}$$

We have obtained a bounded lattice in which $n(x)$ is a complement of x.

3.4 Distributivity from splitting

A complemented lattice need not be distributive. The additional information we need comes from (S) at $(x, y, z) = (a, c, a)$:

$$(a * c) * (c * n(a)) = c.$$

In our lattice notation this is the splitting identity

$$\boxed{c = (c \wedge a) \vee (c \wedge n(a)).} \tag{15}$$

Indeed, expanding the right-hand side as NAND terms and applying double negation gives $(c * a) * (c * n(a))$.

Splitting first yields the useful criterion

$$u \leq v \quad \Longleftrightarrow \quad u \wedge n(v) = \mathbf{0}. \tag{16}$$

For the forward direction, monotonicity of meet gives $u \wedge n(v) \leq v \wedge n(v) = \mathbf{0}$. For the reverse direction, (15) gives

$$u = (u \wedge v) \vee (u \wedge n(v)) = u \wedge v \leq v.$$

Now set

$$t = x \wedge (y \vee z), \qquad d = (x \wedge y) \vee (x \wedge z).$$

Both $x \wedge y$ and $x \wedge z$ lie below t, so $d \leq t$. To prove the other inequality, consider $r = t \wedge n(d)$. We will show $r = \mathbf{0}$, which implies $t \leq d$ by (16).

We have $r \leq x$ and $r \leq n(d)$. Since $x \wedge y \leq d$, order reversal gives $r \leq n(x \wedge y)$. Consequently,

$$r \wedge y \leq x \wedge y, \qquad r \wedge y \leq r \leq n(x \wedge y).$$

Thus $r \wedge y$ is below the meet of an element and its complement, so $r \wedge y = \mathbf{0}$. By (16) and double negation, $r \leq n(y)$. The same argument with z gives $r \leq n(z)$. Using (13),

$$r \leq n(y) \wedge n(z) = n(y \vee z).$$

But also $r \le t \le y \vee z$. Therefore r lies below an element and its complement, so $r = \mathbf{0}$, as desired. We have proved

$$x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z). \tag{17}$$

Applying n to (17) with x, y, z replaced by $n(x), n(y), n(z)$, and using De Morgan's laws, gives the dual distributive law.

It follows that $(A, \wedge, \vee, n, \mathbf{0}, \mathbf{1})$ is a Boolean algebra. Moreover,

$$n(x \wedge y) = n(n(x * y)) = x * y,$$

so its NAND operation is exactly the operation with which we started.

4 Converse and equational completeness

4.1 Checking the axiom in a Boolean algebra

In any Boolean algebra, let $a * b = \neg(a \wedge b)$. The ordinary Boolean laws give

$$(a * b) * c = (a \wedge b) \vee \neg c, \qquad a * ((a * c) * a) = \neg a \vee \neg c.$$

For the second equality, one can first calculate $(a * c) * a = (a \wedge c) \vee \neg a$, and then take NAND with a. It follows that

$$\begin{aligned} &((a * b) * c) * (a * ((a * c) * a)) \\ &\quad = \neg\big(((a \wedge b) \vee \neg c) \wedge (\neg a \vee \neg c)\big) \\ &\quad = \neg\big(\neg c \vee (a \wedge b \wedge \neg a)\big) \\ &\quad = c. \end{aligned}$$

Thus (W) holds, completing the proof of Theorem 1. □

4.2 Why every Boolean NAND identity follows

The model characterization also gives the claimed completeness. We spell out the short quotient argument so that this last step does not rely on an unstated completeness theorem.

Let T be the set of terms formed from variables $x_0, x_1, \ldots$ and a single binary operation symbol $*$. Write $s \equiv t$ if the equation $s = t$ can be derived using substitution instances of (W), reflexivity, symmetry, transitivity, and replacement of equal subterms. This is an equivalence relation and a congruence:

$$s \equiv s', \quad t \equiv t' \quad \Longrightarrow \quad s * t \equiv s' * t'.$$

Thus the quotient $F = T/\!\equiv$ has a well-defined operation $[s] * [t] = [s * t]$. It is nonempty, since it contains $[x_0]$. All substitution instances of (W) are among the generating relations, so F satisfies (W). By Theorem 1, F is the NAND reduct of a Boolean algebra.

Suppose now that $s = t$ is valid under every interpretation in every Boolean algebra. Interpret each variable x_i as $[x_i] \in F$. Induction on terms shows that a term r then evaluates to $[r]$. Validity therefore gives $[s] = [t]$, which, by definition of the quotient, means $s \equiv t$. The identity is derivable from (W). Conversely, every derivable identity is valid in every Boolean algebra, because the axiom and all equality rules are valid there. We conclude:

Theorem 3 (Equational completeness)**.** *An identity in the NAND language holds in every Boolean algebra if and only if it is derivable from* (W) *by equational reasoning.*